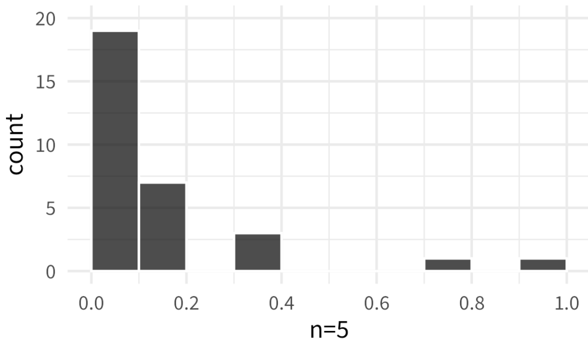
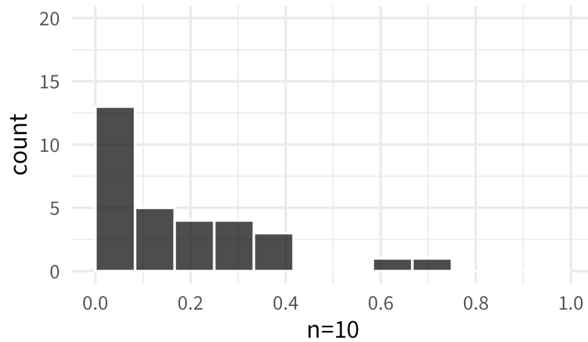
NOTES 10 ADDENDUM: FINGERPRINTS SAMPLING DISTRIBUTIONS

Stat 120 | Spring 2026

Prof Amanda Luby

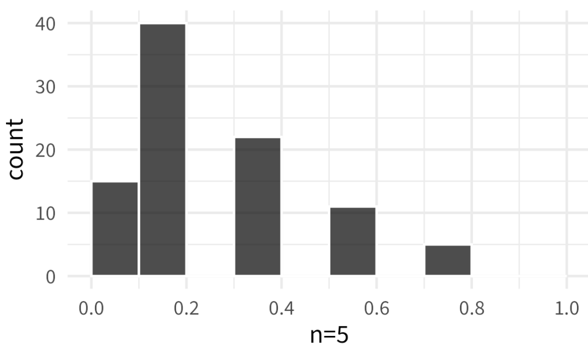
Non-random samples: (from our class)

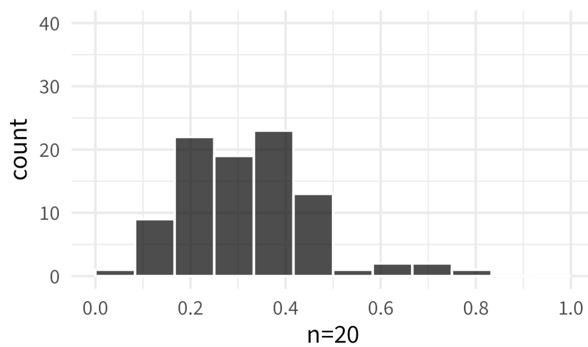




- As n increases, there is less spread in the sampling distribution
- Because our samples are biased (our classroom is not a random sample of the population), there's no guarantee that our distribution is getting closer to the "right answer" (p_{whorl})

Random samples: (from fingerprint generator)





- As n increases, the sampling distribution becomes more symmetric and less spread out
- As n increases, the "peak" of our distribution gets closer and closer to the true parameter
- Shows us the expected **sampling variability** of the sample statistic for a given sample size